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submitted by

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Analysis of Cosmic Structure Formation under the Lens of Zel'dovich Flow and the Method of Characteristics

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Abstract

This thesis investigates cosmic structure formation using the Zel'dovich approximation and the method of characteristics, emphasizing the evolution of density perturbations and the onset of shell-crossing in a Lagrangian framework.

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Conclusion **43**

Motivation

> "Two things fill the mind with ever new and increasing admiration and awe, the more often and steadily we reflect upon them: the starry heavens above me and the moral law within me."
> — Immanuel Kant, *Critique of Practical Reason*

CHAPTER 1

Introduction

This thesis can broadly be split into three sections: Fourier analysis of the Vlasov equation under a perturbative Zel'dovich split. The introduction of the method of characteristics as an approach to solving said Vlasov equation, followed by the analysis of how the mass probability function changes when subjected to higher order gravitational contributions under Born approximation.

CHAPTER 2

Mathematical and Theoretical Background

2.1 Cosmological Foundations

2.1.1 Cosmological Background Dynamics

The cosmological standard model rests on three foundational assumptions: that general relativity adequately describes gravity on cosmological scales, that the universe appears statistically isotropic, and that this isotropy holds for all freely falling observers (the Copernican principle). Since the strong and weak interactions are confined to subatomic scales, and the electromagnetic interaction can be screened, gravity is the only force that shapes the large-scale dynamics of the cosmos [1]. Imposing spatial homogeneity and isotropy restricts the space-time metric to the Robertson–Walker form, whose single dynamical degree of freedom is the cosmic scale factor $a(t)$.

Inserting this metric into Einstein's field equations yields the two Friedmann equations. Following [1], combining them gives the first-order Friedmann equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\,\rho - \frac{Kc^2}{3a^2} + \frac{\Lambda c^2}{3}\,, \tag{2.1}$$

together with the adiabatic equation

$$\mathrm{d}\Big(\rho\, c^2\, a^3\Big) + P\,\mathrm{d}\Big(a^3\Big) = 0\,, \tag{2.2}$$

which expresses energy conservation in the absence of heat flows—a direct consequence of spatial isotropy. Here ρ is the mass–energy density, P the pressure, K the spatial curvature, and Λ the cosmological constant. Measurements are consistent with vanishing spatial curvature, so we set $K = 0$ henceforth.

For non-relativistic matter ("dust", $P = 0$) and relativistic matter ("radiation", $P = \rho c^2/3$), the adiabatic equation implies the familiar density scalings $\rho_\mathrm{m} \propto a^{-3}$ and $\rho_\mathrm{r} \propto a^{-4}$, respectively.

Defining the Hubble function $H \equiv \dot{a}/a$ and the present-day density parameters

$$\Omega_{\mathrm{m0}} = \frac{8\pi G\,\rho_{\mathrm{m0}}}{3H_0^2}\,, \qquad \Omega_{\mathrm{r0}} = \frac{8\pi G\,\rho_{\mathrm{r0}}}{3H_0^2}\,, \qquad \Omega_{\Lambda 0} = \frac{\Lambda c^2}{3H_0^2}\,, \tag{2.3}$$

the Friedmann equation can be cast into the compact form

$$E^2(a) \equiv \frac{H^2}{H_0^2} = \Omega_{\mathrm{r0}}\,a^{-4} + \Omega_{\mathrm{m0}}\,a^{-3} + \Omega_{\Lambda 0}\,. \tag{2.4}$$

2.1.2 Comoving Coordinates and Linear Structure Growth

To separate the overall cosmic expansion from the dynamics of density perturbations, one introduces comoving coordinates $\boldsymbol{q}$ related to physical coordinates by $\boldsymbol{x} = a\,\boldsymbol{q}$. The velocity field then decomposes into the Hubble flow $\boldsymbol{v}_{\mathrm{h}} = H\,\boldsymbol{x}$ and a peculiar velocity $\boldsymbol{u}$ relative to it,

$$\boldsymbol{v} = \dot{\boldsymbol{x}} = \dot{a}\,\boldsymbol{q} + a\,\dot{\boldsymbol{q}} = \boldsymbol{v}_{\mathrm{h}} + a\,\boldsymbol{u}\,. \tag{2.5}$$

Rewriting the continuity and Euler equations in comoving coordinates, introducing the dimensionless density contrast $\delta = (\rho - \bar{\rho})/\bar{\rho}$, and subtracting the homogeneous background contributions, one arrives at the perturbed Euler–Poisson system for pressureless matter (see [1], Sec. 1.2 for the full derivation):

$$\begin{aligned} \partial_t\delta + \nabla\cdot\Big[(1+\delta)\,\boldsymbol{u}\Big] &= 0\,, \\ \partial_t\boldsymbol{u} + 2H\,\boldsymbol{u} + \Big(\boldsymbol{u}\cdot\nabla\Big)\boldsymbol{u} &= -a^{-2}\,\nabla\Phi\,, \\ \nabla^2\Phi &= 4\pi G\,\bar{\rho}\,a^2\,\delta\,. \end{aligned} \tag{2.6}$$

Here the Poisson equation contains only the density fluctuation δ as its source, not the full density—a procedure justified rigorously through general relativity but often introduced in the Newtonian framework as the Jeans swindle [1].
Linearising this system by dropping all terms beyond first order in $(\delta, \boldsymbol{u})$, and eliminating the peculiar velocity, yields the linear growth equation

$$\ddot{\delta} + 2H\,\dot{\delta} = 4\pi G\,\bar{\rho}\,\delta\,. \tag{2.7}$$

Its growing-mode solution $D_+(a)$ defines the linear growth factor, which quantifies how density perturbations amplify with cosmic time. In an Einstein–de Sitter (EdS) universe, where $\Omega_{\mathrm{m}} = 1$ at all times, the growth factor is simply $D_+ = a$. More generally, $D_+(a)$ must be obtained by solving Eq. (2.7) numerically with the full expansion function (2.4). The growth factor will serve as a convenient time coordinate in what follows.

2.1.3 Particle Trajectories in the Expanding Background

Rather than treating cosmic structure formation in the hydrodynamical picture, one can equivalently follow the trajectories of individual dark-matter particles. In comoving coordinates $\boldsymbol{q}$, the Lagrangian for a test particle takes the form [1]

$$\tilde{L} = \frac{a^2}{2}\,\dot{\boldsymbol{q}}^{\,2} - \varphi\,, \tag{2.8}$$

where the potential φ satisfies $\nabla^2\varphi = 4\pi G\,\bar{\rho}\,\delta\,a^2$. This form is obtained from the physical Lagrangian by a gauge transformation that absorbs the homogeneous background dynamics. Crucially, the second Friedmann equation eliminates the mean density and the cosmological constant from the Poisson equation, leaving only the density fluctuation δ as the source—providing a clean derivation of what was earlier introduced as the Jeans swindle (cf. Sec. 2.1.2). Introducing the shifted linear growth factor $\tau = D_+ - D_+^{(i)}$ as a new time coordinate and defining the effective, time-dependent particle mass

$$m(\tau) = a^3\,E\,D'_+\,, \tag{2.9}$$

where primes denote derivatives with respect to a, the Lagrangian assumes its final form [1]

$$L = \frac{m}{2}\,\dot{\boldsymbol{q}}^{\,2} - m\,\phi\,, \tag{2.10}$$

where the rescaled potential ϕ satisfies $\nabla^2\phi = A_\phi\,\delta$ with $A_\phi = 3a/(2m^2)$, and dots now denote derivatives with respect to τ. In the EdS limit, $m = a^{3/2}$.
The resulting Hamiltonian equations can be solved formally through variation of constants. Writing the phase-space position as $\mathsf{x} = (\boldsymbol{q}, \boldsymbol{p})$, the full trajectory reads [1]

$$\mathsf{x}(\tau) = \mathsf{G}(\tau,0)\,\mathsf{x}^{(i)} + \int_0^\tau \mathrm{d}\tau'\;\mathsf{G}(\tau,\tau')\begin{pmatrix}\mathbf{0}\\ -m\,\nabla\phi\end{pmatrix}, \tag{2.11}$$

with the Green's function

$$\mathsf{G}(\tau,\tau') = \begin{pmatrix}\mathbf{1}_3 & g_{\mathrm{H}}(\tau,\tau')\,\mathbf{1}_3\\ \mathbf{0} & \mathbf{1}_3\end{pmatrix} \tag{2.12}$$

built from the Hamiltonian propagator $g_{\mathrm{H}}(\tau,\tau') = \int_{\tau'}^{\tau}\mathrm{d}\bar{\tau}\,/\,m(\bar{\tau})$. This integral equation is exact but must be solved self-consistently, since the potential gradient on the right-hand side must be evaluated along the trajectory itself. For the detailed derivation of the gauge transformation, the time-coordinate change, and the variation-of-constants solution, we refer to [1], Secs. 2.1–2.2.

2.1.4 Zel'dovich Trajectories

Equation (2.11) suggests an iterative solution starting from the free (force-free) Hamiltonian motion. However, the Hamiltonian propagator g_{H} is bounded and small—for example, $g_{\mathrm{H}} \leq 2$ in the EdS model [1]—meaning that freely streaming particles barely move in comoving coordinates due to the expansion drag encoded in the time-dependent mass. As a consequence, the integral term in Eq. (2.11) is generically large, and a perturbative expansion around Hamiltonian free-streaming converges slowly.
The key idea behind the Zel'dovich approximation is to choose a different set of reference trajectories that already incorporate the dominant gravitational effect—namely linear structure growth. Consider particles streaming along

$$\boldsymbol{q}(\tau) = \boldsymbol{q}^{(i)} + \tau\,\boldsymbol{u}^{(i)}\,, \tag{2.13}$$

where $\boldsymbol{u}^{(i)}$ is the initial peculiar velocity. If the initial velocity field is irrotational, $\boldsymbol{u}^{(i)} = \nabla\psi$ for some velocity potential ψ, the Jacobian of the mapping from initial to final positions is

$$J_{ij} = \frac{\partial q_i}{\partial q_j^{(i)}} = \delta_{ij} + \tau\,\partial_i\partial_j\,\psi\,. \tag{2.14}$$

Mass conservation then implies $\rho = \rho^{(i)}/\det \mathsf{J}$. Expanding the determinant to first order at early times and using the linearised continuity equation $\nabla^2\psi = -\delta^{(i)}$, one finds

$$\delta(\tau) \approx (1+\tau)\,\delta^{(i)} = D_+\,\delta^{(i)}\,, \tag{2.15}$$

which correctly reproduces the linear growth of density perturbations—without any explicit gravitational interaction. This is possible because the initial velocities are not independent of the density field; they are related to it through the Poisson equation, as reflected in the linearised continuity equation [1].
Formally, this amounts to replacing the Hamiltonian propagator by the Zel'dovich propagator

$$g_{\mathrm{H}}(\tau,\tau') \;\longrightarrow\; g_{\mathrm{Z}}(\tau,\tau') = \tau - \tau'\,. \tag{2.16}$$

In [1], Sec. 2.2, it is shown that the residual force acting between Zel'dovich trajectories depends only on the *nonlinear* density contrast $\delta^{(\mathrm{nl})}$, obtained by subtracting the linearly evolved contribution $D_+\,\delta^{(i)}$ from the full density contrast. A new potential φ sourced by $\delta^{(\mathrm{nl})}$ alone can be introduced, and the complete phase-space trajectory takes the form

$$\mathbf{x}(\tau) = \mathsf{G}_{\mathrm{Z}}(\tau,0)\,\mathbf{x}^{(i)} + \int_0^\tau \mathrm{d}\tau'\;\mathsf{G}_{\mathrm{Z}}(\tau,\tau')\begin{pmatrix}\mathbf{0}\\ \boldsymbol{\mathcal{F}}\end{pmatrix}, \tag{2.17}$$

with the Zel'dovich Green's function

$$\mathsf{G}_{\mathrm{Z}}(\tau,\tau') = \begin{pmatrix}\mathbf{1}_3 & (\tau-\tau')\,\mathbf{1}_3\\ \mathbf{0} & \mathbf{1}_3\end{pmatrix}, \tag{2.18}$$

and an effective force $\boldsymbol{\mathcal{F}}$ that is sourced only by nonlinear density fluctuations.
Since the nonlinear density contrast vanishes at early times, the free Zel'dovich trajectories (2.13) provide an excellent approximation at sufficiently early epochs and offer a natural starting point for perturbative corrections. This is the decisive advantage of the Zel'dovich frame over the Hamiltonian frame: it absorbs the gravitational effects of linear structure formation into the reference motion, so that only the genuinely nonlinear dynamics remain to be treated perturbatively [1, 2].

2.2 Klimontovich Equation and the BBGKY Hierarchy

We consider a classical dynamical system composed of N identical point-like particles. At any given time t, the exact microstate of the system is entirely specified by the set $\{x_j\}_{j=1}^N$ of particle phase-space coordinates $x_j = (\vec{q}_j, \vec{p}_j)$, which together constitute a point in the $6N$-dimensional phase space Γ. The time evolution of the system is then given by a trajectory $x(t)$ in phase space. Thus, in order to understand the time evolution of the system as a whole we need to understand the dynamical laws governing the motion of the individual particles. In the Hamiltonian description of classical mechanics these dynamics are fully determined by the (possibly time-dependent) Hamiltonian function $H(x,t)$, which, in many physical systems, corresponds to the total energy. More precisely, the individual trajectories are subject to the Hamiltonian equations of motion [2, 3]

$$\frac{d}{dt}\vec{q}_j(t) = \vec{\nabla}_{\vec{p}_j} H(x_j,t), \qquad \frac{d}{dt}\vec{p}_j(t) = -\vec{\nabla}_{\vec{q}_j} H(x_j,t), \qquad j = 1,\dots,N \tag{2.1}$$

which can be written more compactly by introducing the symplectic matrix ω as

$$\dot{x}_j(t) = \omega\, \nabla_{x_j} H\Big(x_j(t), t\Big), \qquad \omega = \begin{pmatrix} 0 & I_{3\times3} \\ -I_{3\times3} & 0 \end{pmatrix} \tag{2.2}$$

The N-particle Hamiltonian function can be decomposed into a free part H_0 and an interacting part H_{int} as [2]

$$H(x,t) = H_0 + H_{\text{int}}(x,t) \tag{2.3}$$

$$= \frac{1}{2} x^{\mathrm{T}} A(t) x + \frac{1}{2} \sum_{i=1}^{N} v_1(x_i, t) + \frac{1}{2} \sum_{\substack{i,j=1 \\ i\neq j}}^{N} v_2(x_i, x_j, t) \tag{2.4}$$

where $A(t)$ is some symmetric matrix encoding the free dynamics, and the v_1 term represents external field contributions and v_2 pairwise interactions. The possibly time-dependent two-particle potential $v_2(x_i, x_j, t)$ governs the pairwise interactions between particles; in the cosmological context this will be the Newtonian gravitational interaction potential.
In the absence of interactions, each particle follows the free equations of motion

$$\dot{x}^{(0)}(t) = \omega \cdot A(t) \cdot x^{(0)}(t) \tag{2.5}$$

whose solution defines the free classical trajectory

$$x_{\text{cl}}^{(0)}(t) = G(t, t^{(i)}) \cdot x^{(i)} \tag{2.6}$$

propagated by the free Green's function

$$G(t,t') = \begin{pmatrix} g_{qq}(t,t') I_{3\times3} & g_{qp}(t,t') I_{3\times3} \\ g_{pq}(t,t') I_{3\times3} & g_{pp}(t,t') I_{3\times3} \end{pmatrix} \Theta(t-t') \tag{2.7}$$

where the Heaviside function $\Theta(t-t')$ ensures causality, and the scalar functions g_{qq}, g_{qp}, g_{pq}, g_{pp} encode the specific structure of the free Hamiltonian.
For systems with a very large number of particles, $N \gg 1$, tracking every individual trajectory is neither practical nor necessary. Instead, one is interested in collective phenomena such as density correlations. This motivates a statistical description through the *Klimontovich phase-space density* [3, 2], which represents the exact microscopic particle density in the six-dimensional single-particle phase space (μ-space):

$$\Phi_f(x,t) = \sum_{i=1}^{N} \delta_D(x - x_i(t)) \tag{2.8}$$

This is a sum of Dirac delta distributions peaked at the instantaneous positions of all N particles. It encodes the full microscopic information of the system in μ-space, in contrast to the $6N$-dimensional Liouville phase-space density $\varrho_N(x,t)$ which lives in the full Γ-space.
The ensemble-averaged Klimontovich density defines the one-particle reduced distribution function [3, 2]

$$f_1(x,t) = \langle \Phi_f(x,t) \rangle = \int dx \sum_{i=1}^{N} \delta_D(x - x_i(t)) \rho_N(x,t) \tag{2.9}$$

which describes the probability of finding any particle at phase-space coordinate x at time t, and is normalized to the total particle number:

$$\int d^6x\, f_1(x,t) = N \tag{2.10}$$

The two-point correlation function of the Klimontovich density decomposes into a self-correlation (shot-noise) contribution and a smooth two-particle density [2]:

$$\bar{C}_{ff}(x_1,x_2,t) = \langle \Phi_f(x_1,t)\Phi_f(x_2,t)\rangle = \delta_D(x_1-x_2)f_1(x_1,t) + f_2(x_1,x_2,t) \tag{2.11}$$

The first term arises from the self-pairing of individual particles ($i = j$ in the double sum) and reflects the discrete, granular nature of the particle distribution. The two-particle reduced distribution function f_2 can further be split into a product of one-particle densities and an irreducible two-particle correlation:

$$f_2(x_1,x_2,t) = f_1(x_1,t)f_1(x_2,t) + g_2(x_1,x_2,t) \tag{2.12}$$

where g_2 measures the genuine statistical correlations between two particles beyond what is already contained in their individual distributions. If the particles were completely uncorrelated, g_2 would vanish and f_2 would factorize.
The connected two-point correlator is obtained by subtracting the disconnected part:

$$\bar{G}_{ff}(x_1,x_2,t) = \langle \Phi_f(x_1,t)\Phi_f(x_2,t)\rangle_c = \delta_D(x_1-x_2)f_1(x_1,t) + g_2(x_1,x_2,t) \tag{2.13}$$

Even for an uncorrelated state where $g_2 = 0$, the connected correlator retains a shot-noise contribution from the self-correlation of individual particles.
This construction generalizes to higher orders. The s-particle reduced distribution function is defined as [2, 3]

$$f_s(x_1,\dots,x_s,t) := \left\langle \sum_{\substack{i_1,\dots,i_s=1\\ i_1\neq\cdots\neq i_s}} \prod_{n=1}^{s} \delta_D(x_n - x_{i_n}(t)) \right\rangle \tag{2.14}$$

where the restriction $i_1 \neq \cdots \neq i_s$ ensures that distinct particles are counted. It satisfies the normalization

$$\int d^6x_1 \cdots d^6x_s\, f_s(x_1,\dots,x_s,t) = \frac{N!}{(N-s)!} \tag{2.15}$$

reflecting the number of ordered selections of s particles from N. For a completely uncorrelated (ideal) system, the s-particle distribution factorizes into a product of one-particle densities:

$$f_s(x_1,\dots,x_s,t) = \prod_{n=1}^{s} f_1(x_n,t) \tag{2.16}$$

Similarly, the k-point correlation function of the Klimontovich density reads

$$\bar{C}_{f\dots f}(x_1,\dots,x_k,t) := \langle \Phi_f(x_1,t)\cdots\Phi_f(x_k,t)\rangle \tag{2.17}$$

$$= \int dx \prod_{i=1}^{k}\sum_{i_n=1}^{N} \delta_D(x_i - x_i(t))\rho_N(x,t) \tag{2.18}$$

which contains both the connected (irreducible) correlations and the disconnected (reducible) contributions that factorize into lower-order correlators.
We now turn to the dynamical evolution. Since the time dependence of the Klimontovich density $\Phi_f(x,t)$ enters only implicitly through the particle trajectories, we can compute its time derivative using the chain rule and the Hamiltonian equations of motion [2]:

$$\frac{\partial}{\partial t_1}\Phi_f(x_1,t_1) = \sum_{i=1}^{N} \nabla_{x_i}^T \delta_D(x_1 - x_i(t)) \cdot \omega \cdot \nabla_{x_i} H(x,t) \tag{2.19}$$

Inserting the decomposition of the Hamiltonian (2.4) and replacing the particle coordinates by μ-space coordinates using the Dirac delta distributions of Φ_f, one arrives at the *Klimontovich equation* [3, 2]:

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1) - \hat{\mathcal{L}}_1^{(1)}(t_1)\right]\Phi_f(x_1,t_1) = \int d^6x_2 \hat{\mathcal{L}}_{12}^{(2)}(t_1)\Phi_f(x_1,t_1)\Phi_f(x_2,t_1) \tag{2.20}$$

where we have identified three distinct operators:

$$\hat{\mathcal{L}}_1^{(0)}(t_1) = x_1^T \cdot A(t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.21}$$

$$\hat{\mathcal{L}}_1^{(1)}(t_1) = \nabla_{x_1}^T v_1(x_1,t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.22}$$

$$\hat{\mathcal{L}}_{12}^{(2)}(t_1) = \nabla_{x_1}^T v_2(x_1,x_2,t_1) \cdot \omega \cdot \nabla_{x_1} \tag{2.23}$$

The free evolution operator $\hat{\mathcal{L}}_1^{(0)}$ generates the Hamiltonian flow associated with the free Hamiltonian. The one-particle interaction operator $\hat{\mathcal{L}}_1^{(1)}$ accounts for the influence of external fields acting independently on each particle. Both of these are linear in Φ_f and do not couple phase-space densities at different positions, hence they do not generate correlations. The two-particle interaction operator $\hat{\mathcal{L}}_{12}^{(2)}$ on the right-hand side introduces a nonlinear coupling: the density at position x_2 sources an interaction potential $v_2(x_1,x_2,t_1)$ that affects the evolution of the density at x_1. This term is responsible for the generation of correlations between particles through their mutual interactions.
The Klimontovich equation is an exact representation of the microscopic dynamics in μ-space and contains the same information as the N-particle Liouville equation [3, 2].
To extract the macroscopic content, we take the ensemble average of the Klimontovich equation. This yields the first equation of the BBGKY hierarchy (named after Bogoliubov, Born, Green, Kirkwood, and Yvon), which governs the evolution of the one-particle distribution f_1. However, this equation is not closed: the two-particle interaction operator couples f_1 to the two-particle distribution f_2. In turn, the evolution of f_2 depends on the three-particle distribution f_3, and so forth. This recursive structure gives rise to the BBGKY hierarchy [3, 2], whose general s-th member reads

$$\begin{aligned}\left[\frac{\partial}{\partial t_1} - \sum_{i=1}^{s}\left(\hat{\mathcal{L}}_i^{(0)}(t_1) + \hat{\mathcal{L}}_i^{(1)}(t_1)\right)\right] f_s(x_1,\ldots,x_s,t_1) \\ = \int d^6x_{s+1} \sum_{i=1}^{s} \hat{\mathcal{L}}_{i,s+1}^{(2)}(t_1) f_{s+1}(x_1,\ldots,x_{s+1},t_1)\end{aligned} \tag{2.24}$$

The left-hand side describes the free streaming of s particles under the influence of external fields. The right-hand side, often referred to as the *collision integral*, encodes the effect of

pairwise interactions between any of the s measured particles and an additional particle at x_{s+1} that is integrated over. As the BBGKY hierarchy is exact and retains the full complexity of the Liouville equation, it only closes after the N-th iteration. The central task of kinetic theory is therefore to find systematic truncation procedures that reduce the hierarchy to a closed set of equations involving only the lowest few distribution functions [3].

2.3 Vlasov Equation

In the BBGKY Hierarchy, we notice that knowledge on the evolution of the s particle distribution function requires knowledge of the $(s+1)$ particle distribution function. To close this hierarchy, we must find a way to truncate the equation in a way that we can solve it [3, 2].
We can take the free streaming ansatz, which assumes no pairwise interactions between particles:

$$v_2(x_1, x_2, t) \to \lambda v_2(x_1, x_2, t) \tag{2.1}$$
$$f_1(x_1, t) \sim \Theta(\lambda^{(0)}) \tag{2.2}$$
$$g_2(x_1, x_2, t) \sim \Theta(\lambda^{(1)}) \tag{2.3}$$

we set $\lambda \to 0$ and neglect all terms of order $\Theta(\lambda^{(1)})$ and higher, yielding:

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_i^{(0)}(t_1)\right] f_1(x_1, t_1) = 0 \tag{2.4}$$
$$f_1(x_1, t_1 = t^{(i)}) =: f^{(i)}(x_1) \tag{2.5}$$

we introduce the retarded Green's function [2]:

$$D_{0,R}(x_1, t_1, x_2, t_2) = \delta_D(x_1 - G(t_1, t_2)x_2)\Theta(t_1 - t_2) \tag{2.6}$$
$$\Rightarrow \left[\partial_{t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1)\right] D_{0,R}(x_1, t_1, x_2, t_2) = \delta_D(x_1 - x_2)\delta_D(t_1 - t_2) \tag{2.7}$$
$$\tag{2.8}$$

from which it follows:

$$f_1(x_1, t) = \int d^6x_2 D_{0,R}(x_1, t, x_2, t^{(i)}) f^{(i)}(x_2) \tag{2.9}$$
$$= f^{(i)}(G^{-1}(t, t^{(i)})x_1)\Theta(t_1 - t^{(i)}) \tag{2.10}$$

including the next order in coupling leads to the Vlasov equation [3, 2]:

$$\left[\frac{\partial}{\partial t_1} - \hat{\mathcal{L}}_1^{(0)}(t_1) - \hat{\mathcal{L}}_1^{(1)}(t_1)\right] f_1(x_1, t_1) = \int d^6x_2 \hat{\mathcal{L}}_{12}^{(2)}(t_1) f_1(x_1, t_1) f_1(x_2, t_1) \tag{2.11}$$

2.3.1 Einstein De Sitter Universe

Start with conformal time τ and comoving coords $\mathbf{x}$: $\mathbf{p} = am\,\frac{d\mathbf{x}}{d\tau} \implies \frac{d\mathbf{x}}{d\tau} = \frac{\mathbf{p}}{am}$.

$$\frac{df}{d\tau} = 0 \implies \frac{\partial f}{\partial \tau} + \frac{\partial f}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{d\tau} + \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{d\mathbf{p}}{d\tau} = 0 \tag{2.12}$$

Since we are working with the canonical momentum, we can extract the first derivative of the momentum from the Hamilton equation:

$$\dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}}, \quad \text{where } q = x \text{ and } \mathcal{H}(p,x) = \frac{p^2}{2m} + \underbrace{V(x)}_{=\tilde{\Phi}} \tag{2.13}$$

$$\implies \frac{d\mathbf{p}}{d\tau} = -am\,\nabla_x\tilde{\Phi} \implies \frac{df}{d\tau} = \frac{\partial f}{\partial \tau} + \frac{1}{am}\nabla_x f - am\,\nabla_x\tilde{\Phi}\,\nabla_p f = 0 \tag{2.14}$$

where

$$\Delta\tilde{\Phi}(\mathbf{x},\tau) = \frac{3}{2}\,\Omega_m(\tau)\,H^2(\tau)\,\delta(\mathbf{x},\tau) \tag{2.15}$$

Furthermore in comoving coords we have [1]:

$$\ddot{\mathbf{q}} + H\dot{\mathbf{q}} = -\nabla_{\mathbf{q}}\tilde{\Phi} \tag{2.16}$$

$\implies$ we introduce $\eta = \log\left(\frac{D(\tau)}{D^{(i)}}\right)$ and the non-canonical momentum $\mathbf{p} = m\,\frac{d\mathbf{x}}{d\eta}$.

$$\implies \frac{df}{d\eta} = \frac{\partial f}{\partial \eta} + \frac{\partial f}{\partial \mathbf{x}}\cdot\frac{d\mathbf{x}}{d\eta} + \frac{\partial f}{\partial \mathbf{p}}\cdot\frac{d\mathbf{p}}{d\eta} = 0 \tag{2.17}$$

$$\boxed{\begin{aligned} &f_+ = \tfrac{d\log b_+}{d\log a},\ H = \tfrac{\dot{a}}{a};\ \eta = \log\left(\tfrac{D(\tau)}{D^{(i)}}\right) \\ &= \tfrac{\dot{D}_+}{D_+}\cdot\tfrac{a}{\dot{a}} = \tfrac{d\eta}{d\tau}\cdot\tfrac{1}{H} \\ &\implies \tfrac{d\eta}{d\tau} = f_+ H \end{aligned}}$$

$$\dot{\mathbf{q}} = \dot{\mathbf{x}} = \frac{d\mathbf{x}}{d\tau} = \frac{\partial \mathbf{x}}{\partial \eta}\cdot\frac{d\eta}{d\tau} = \frac{\mathbf{p}}{m}\cdot f_+ H \tag{2.18}$$

$$\ddot{\mathbf{q}} = \frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] \tag{2.19}$$

$$= \frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] \tag{2.20}$$

From the linear growth equation (2.7):

$$\ddot{D} + H\dot{D} = \frac{3}{2}\Omega_m H^2 D \implies \frac{\ddot{D}}{D} + H\frac{\dot{D}}{D} = \frac{3}{2}\Omega_m H^2 \tag{2.21}$$

$$\implies \frac{d}{d\tau}(f_+H) = \frac{\ddot{D}}{D} - \left(\frac{\dot{D}}{D}\right)^2 = \frac{3}{2}\Omega_m H^2 - f_+H^2 - (f_+H)^2 \tag{2.22}$$

Inserting into (2.16) we get

$$\frac{1}{m}\left[(f_+H)^2\frac{d\mathbf{p}}{d\eta} + \mathbf{p}\frac{d}{d\tau}(f_+H)\right] + \frac{\mathbf{p}}{m}f_+H^2 = -\nabla_{\mathbf{q}}\tilde{\Phi} \tag{2.23}$$

$$\implies \frac{d\mathbf{p}}{d\eta} = -\mathbf{p}\left[\frac{3}{2}\frac{\Omega_m}{f_+} - 1\right] - \nabla_{\mathbf{q}}\tilde{\Phi}\cdot m \approx -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi} \tag{2.24}$$

using the EdS approximation: $\Omega_m = f_+ = 1$, and $\tilde{\Phi} = \Phi/(Hf_+)^2$.

$$\boxed{\frac{\partial f}{\partial \eta} + \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{q}} f - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}\right]\cdot\nabla_{\mathbf{p}} f = 0} \tag{2.25}$$

2.4 Momentum Moments and Closure

Let $f_s(\mathbf{q},\mathbf{p},\tau)$ be the full 6D phase-space density. We define the momentum moments [2]

$$n(\mathbf{q},\tau) := \int d\mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.1}$$

$$\Pi(\mathbf{q},\tau) := \int d\mathbf{p}\; \mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau) = am\, n(\mathbf{q},\tau)\; \mathbf{u}(\mathbf{q},\tau), \tag{2.2}$$

$$T(\mathbf{q},\tau) := \int d\mathbf{p}\; \mathbf{p}\otimes\mathbf{p}\; f_s(\mathbf{q},\mathbf{p},\tau) = a^2m^2 n(\mathbf{q},\tau)\Big(\mathbf{u}(\mathbf{q},\tau)\otimes\mathbf{u}(\mathbf{q},\tau)+\sigma(\mathbf{q},\tau)\Big), \tag{2.3}$$

where the bulk velocity and velocity-dispersion tensor are

$$u_i(\mathbf{q},\tau) = \frac{1}{am\, n(\mathbf{q},\tau)}\int d\mathbf{p}\; p_i\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.4}$$

$$\sigma_{ij}(\mathbf{q},\tau) = \frac{1}{a^2m^2 n(\mathbf{q},\tau)}\int d\mathbf{p}\; \Big(p_i - am\, u_i(\mathbf{q},\tau)\Big)\Big(p_j - am\, u_j(\mathbf{q},\tau)\Big)\; f_s(\mathbf{q},\mathbf{p},\tau). \tag{2.5}$$

We describe the new degrees of freedom in a way that avoids redundancy $\rightarrow$ *cumulants*. Define the characteristic (moment-generating) function

$$Z(\mathbf{q},\mathbf{l},\tau) = \int d\mathbf{p}\; e^{i\mathbf{l}\cdot\mathbf{p}}\; f_s(\mathbf{q},\mathbf{p},\tau), \tag{2.6}$$

with moments

$$M^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau) = i^n\, \frac{\partial^n Z(\mathbf{q},\mathbf{l},\tau)}{\partial l_{i_1}\cdots\partial l_{i_n}}\bigg|_{\mathbf{l}=0}, \tag{2.7}$$

so that (expansion around $\mathbf{l}=0$)

$$Z(\mathbf{q},\mathbf{l},\tau) = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!} M^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}. \tag{2.8}$$

The cumulant-generating function and its cumulants are

$$C(\mathbf{q},\mathbf{l},\tau) := \ln Z(\mathbf{q},\mathbf{l},\tau) = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}, \tag{2.9}$$

$$C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau) = i^n\, \frac{\partial^n C(\mathbf{q},\mathbf{l},\tau)}{\partial l_{i_1}\cdots\partial l_{i_n}}\bigg|_{\mathbf{l}=0}. \tag{2.10}$$

The first three cumulants identify with the physical moments:

$$C^{(0)}(\mathbf{q},\tau) = \ln n(\mathbf{q},\tau), \tag{2.11}$$

$$C^{(1)}_i(\mathbf{q},\tau) = am\, u_i(\mathbf{q},\tau), \tag{2.12}$$

$$C^{(2)}_{ij}(\mathbf{q},\tau) = a^2m^2\, \sigma_{ij}(\mathbf{q},\tau). \tag{2.13}$$

The full distribution can therefore be written as

$$Z(\mathbf{q},\mathbf{l},\tau) = \exp\left(\sum_{n=0}^{\infty}\frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}\right), \tag{2.14}$$

$$f_s(\mathbf{q},\mathbf{p},\tau) = \int \frac{d\mathbf{l}}{(2\pi)^3}\; e^{i\mathbf{l}\cdot\mathbf{p}} \exp\left(\sum_{n=0}^{\infty}\frac{(-i)^n}{n!} C^{(n)}_{i_1\cdots i_n}(\mathbf{q},\tau)\; l_{i_1}\cdots l_{i_n}\right). \tag{2.15}$$

2.4.1 SSA: Single Stream Approximation

- Set all cumulants of orders higher than 1 to zero.
- $f_s(\mathbf{q}, \mathbf{p}, \tau)$ is an infinitely thin 3D sheet in 6D phase space $\rightarrow$ the system is in a perfectly cold state.

$$f_s(\mathbf{q}, \mathbf{p}, \tau) = n(\mathbf{q}, \tau) \int \frac{d\mathbf{l}}{(2\pi)^3} \exp\Big(i\mathbf{l} \cdot (\mathbf{p} - am\, \mathbf{u}(\mathbf{q}, \tau))\Big) = n(\mathbf{q}, \tau)\ \delta^{(3)}\Big(\mathbf{p} - am\, \mathbf{u}(\mathbf{q}, \tau)\Big). \quad (2.16)$$

2.4.2 Gaussian Closure

- Non-vanishing velocity dispersion.
- Set all cumulants of order higher than 2 to zero.

$$\Rightarrow \quad f_s(\mathbf{q}, \mathbf{p}, \tau) = n(\mathbf{q}, \tau) \int \frac{d\mathbf{l}}{(2\pi)^3} \exp\Big(i\mathbf{l} \cdot (\mathbf{p} - am\, \mathbf{u}(\mathbf{q}, \tau)) - \tfrac{1}{2}\sigma_{ij}\, l_i l_j\Big)\,. \quad (2.17)$$

We use the standard Gaussian integral identity: for symmetric, positive definite (and invertible) matrices $A \in \mathbb{R}^{n\times n}$ and $\mathbf{b} \in \mathbb{R}^n$,

$$\int_{\mathbb{R}^n} d^n\mathbf{x}\ \exp\Big(-\tfrac{1}{2}\mathbf{x}^T A\mathbf{x} + \mathbf{b}^T\mathbf{x}\Big) = \frac{(2\pi)^{n/2}}{\sqrt{\det A}}\ \exp\Big(\tfrac{1}{2}\mathbf{b}^T A^{-1}\mathbf{b}\Big)\,. \quad (2.18)$$

The velocity dispersion σ_{ij} is symmetric by definition. We must also require it to be *positive definite*:

$$V_i\, \sigma^{ij}\, V_j = \frac{1}{a^2 m^2 n} \int d\mathbf{p}\ V_i(p_i - am\, u_i)\, V_j(p_j - am\, u_j)\ f_s = \frac{1}{a^2 m^2}\, \mathrm{Var}(\mathbf{V} \cdot \mathbf{p}). \quad (2.19)$$

Positive definiteness requires $V_i \sigma^{ij} V_j > 0$, i.e., $\mathrm{Var}(\mathbf{V} \cdot \mathbf{p}) > 0$: no vanishing support in any direction. This is already guaranteed by the constraint of isotropy. ✓
Identifying $\mathbf{b} = i(\mathbf{p} - am\, \mathbf{u}(\mathbf{q}, \tau))$, we obtain the *Maxwell–Boltzmann form* of the Gaussian closure:

$$\boxed{f_s(\mathbf{q}, \mathbf{p}, \tau) = \frac{n(\mathbf{q}, \tau)}{(2\pi)^{3/2}\sqrt{\det \sigma}} \exp\Big(-\frac{1}{2}(p_i - am\, u_i)(\sigma^{-1})_{ij}(p_j - am\, u_j)\Big)\,.} \quad (2.20)$$

As shown in [4], the Fourier-transformed Vlasov equation under the EdS approximation provides the starting point for the perturbative expansion around the Zel'dovich solution discussed in the following chapter.

CHAPTER 3

Zel'dovich Split

In this section our goal is to solve the Vlasov equation in Fourier space under a perturbative expansion around the Zel'dovich solution. From this section onwards f will be referring to the Fourier transformed one particle distribution function unless otherwise specified.
We start by writing down the Vlasov equation in operator form:

$$\mathcal{L}_0 f + \mathcal{L}_{Grav} f = 0 \tag{3.1}$$

$$\rightarrow \mathcal{L}_0 \left[f_{0,Zel} + \delta f\right] + \mathcal{L}_{Grav} \left[f_{0,Zel} + \delta f\right] = 0 \tag{3.2}$$

We know by construction that the Zel'dovich background $f_{0,Zel}$ follows the Zel'dovich flow, which reproduces linear growth, so by definition the Zel'dovich background satisfies:

$$\mathcal{L}_0 f_{0,Zel} + \mathcal{L}^{(1)}_{Grav} f_{0,Zel} = 0 \tag{3.3}$$

where $\mathcal{L}^{(1)}_{Grav}$ denotes the first order gravitational interaction operator.

$$\mathcal{L}_{Grav} = \mathcal{L}^{(1)}_{Grav} + \mathcal{L}^{(2)}_{Grav} + \dots \tag{3.4}$$

Knowing this we can factor out (3.2) and (3.3), which leaves us with:

$$\mathcal{L}_0 \delta f + \mathcal{L}^{(\text{rest})}_{Grav} \delta f + \mathcal{L}_{Grav} f_{0,Zel} = 0 \tag{3.5}$$

where

$$\mathcal{L}^{(\text{rest})}_{Grav} = \mathcal{L}_{Grav} - \mathcal{L}^{(1)}_{Grav} = \sum_{n \geq 2} \mathcal{L}^{(n)}_{Grav} \tag{3.6}$$

We can further define the Zel'dovich operator as:

$$\mathcal{L}_{Zel} := \mathcal{L}_0 + \mathcal{L}^{(1)}_{Grav} = G_Z^{-1} \tag{3.7}$$

also known as the Zel'dovich 'free' propagator.

$$\mathcal{L}_0\delta f + \mathcal{L}_{Grav}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f \tag{3.8}$$

$$\Leftrightarrow \mathcal{L}_0\delta f + \mathcal{L}^{(1)}_{Grav}\delta f + \sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\delta f \tag{3.9}$$

$$\Leftrightarrow \mathcal{L}_{Zel}\delta f = -\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\left[f_{0,Zel} + \delta f\right] \tag{3.10}$$

From here we can solve for δf using the Zel'dovich propagator/Green's function:

$$\delta f = G_Z(\eta,\eta')\delta f^{(i)}(\eta') - \int d\eta' G_Z(\eta,\eta')\sum_{n\geq 2}\mathcal{L}^{(n)}_{Grav}\left[f_{0,Zel} + \delta f\right] \tag{3.11}$$

while we now have an expression for the perturbations, we still need to explicitly define the higher order gravitational operator. From [4], the Fourier-transformed Vlasov equation under EdS approximation reads as:

$$\left[\partial_\eta - \boldsymbol{k}\cdot\nabla_l + \frac{1}{2}\boldsymbol{l}\cdot\nabla_l\right] f(\boldsymbol{k},\boldsymbol{l},\eta) - \frac{3}{2\bar{\rho}}\int\frac{\mathrm{d}^3k'}{(2\pi)^3}\frac{\boldsymbol{k'}\cdot\boldsymbol{l}}{k'^2}f(\boldsymbol{k}-\boldsymbol{k'},\boldsymbol{l},\eta)\,f(\boldsymbol{k'},0,\eta) = 0 \tag{3.12}$$

Additionally we know the gravitational operator in Fourier space to be [4]:

$$\mathcal{L}_{Grav}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{0,Zel}(k-k',l,\eta) + \delta_Z f(k-k',l,\eta)\right]\left[f_{0,Zel}(k',0,\eta) + \delta_Z f(k',0,\eta)\right] \tag{3.13}$$

we now introduce the $f_{0,Zel} = f_{0,com} + \delta f^{(1)}$, where $(\cdot)^1$ now denotes the first order perturbations with respect to the comoving coordinates and

$$\delta_Z f = \delta f^{(2)} + \delta f^{(3)} + \dots \tag{3.14}$$

furthermore:

$$f_{0,Zel}(k-k',l,\eta) \equiv f_{Z,0} \quad f_{0,Zel}(k',0,\eta) \equiv f^{\Phi}_{Z,0} \tag{3.15}$$

$$\Rightarrow \quad \mathcal{L}_{\mathrm{Grav}}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{Z,0} + \delta_Z f_Z\right]\left[f^{\Phi}_{Z,0} + \delta_Z f^{\Phi}_Z\right] \tag{3.16}$$

$$= -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{Z,0}f^{\Phi}_{Z,0} + f^{\Phi}_{Z,0}\delta_Z f_Z + f_{Z,0}\delta_Z f^{\Phi}_Z + \delta_Z f^{\Phi}_Z\delta_Z f_Z\right] \tag{3.17}$$

furthermore we know that:

$$f_{Z,0} = f_{0,com} + \delta f^{(1)} \tag{3.18}$$

$$\begin{aligned}\mathcal{L}_{\mathrm{Grav}}f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int\frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\Big[&(f_{0,\mathrm{com}} + \delta f^{(1)})(f^{\Phi}_{0,\mathrm{com}} + \delta f^{(1)\Phi}) \\ &+ (f^{\Phi}_{0,\mathrm{com}} + \delta f^{(1)\Phi})\,\delta_Z f_Z + (f_{0,\mathrm{com}} + \delta f^{(1)})\,\delta_Z f^{\Phi}_Z + (\delta_Z f^{\Phi}_Z)(\delta_Z f_Z)\Big]\end{aligned} \tag{3.19}$$

factoring out lets us separate $\mathcal{L}_{Grav}$ in contributions of perturbation, where $f_{0,com}$ is of order 0, $\delta f^{(1)}$ of order 1 and δ_Z denotes perturbations of order 2 and higher.

$$\Rightarrow \quad \mathcal{L}^{(0)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\, f_{0,\text{com}} f^{\Phi}_{0,\text{com}} = 0 \tag{3.20}$$

$$\mathcal{L}^{(1)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\left[f_{0,\text{com}}\,\delta f^{(1)\Phi} + f^{\Phi}_{0,\text{com}}\,\delta f^{(1)}\right] \tag{3.21}$$

$$\sum_{n\geq 2}\mathcal{L}^{(n)}_{\text{Grav}} f = -\frac{3}{2}\frac{1}{\bar{\rho}}\int \frac{d^3k'}{(2\pi)^3}\frac{k'\cdot l}{k'^2}\Big[\delta f^{(1)}\delta f^{(1)\Phi} + f_{0,\text{com}}\,\delta_Z f^{\Phi}_Z + \delta f^{(1)}\delta_Z f^{\Phi}_Z \tag{3.22}$$

$$+ f^{\Phi}_{0,\text{com}}\,\delta_Z f_Z + \delta f^{(1)\Phi}\delta_Z f_Z + \delta_Z f^{\Phi}_Z \delta_Z f_Z\Big] \tag{3.23}$$

Inserting this back into (3.11) yields an expression for the perturbations δf around the Zel'dovich background:

$$\delta f^{(n)}(k,l,\eta) = \int \frac{d^3q_1}{(2\pi)^3}\cdots\frac{d^3q_n}{(2\pi)^3}\delta^{(i)}(q_1)\ldots\delta^{(i)}(q_n)F_n(k,l;q_1,\ldots,q_n,\eta) \tag{3.24}$$

For now we will leave this section as it is and move on to the method of characteristics.

CHAPTER 4

Method of Characteristics

We can also apply the method of characteristics in standard phase space, which will help us more for upcoming sections. In analogy to the previous section we will work ourselves up to the final solution:

Search for trajectories $(\eta(\mathbf{s}), \mathbf{q}(s), \mathbf{p}(s))$ such that

$$\frac{df}{ds}(\eta, \mathbf{q}, \mathbf{p}) = F(f, \eta, \mathbf{q}, \mathbf{p}) \tag{4.1}$$

$$\implies \frac{df}{ds} = \frac{\partial f}{\partial \eta}\frac{d\eta}{ds} + \frac{\partial f}{\partial \mathbf{q}}\frac{d\mathbf{q}}{ds} + \frac{\partial f}{\partial \mathbf{p}}\frac{d\mathbf{p}}{ds} = F \tag{4.2}$$

comparing this form to the Vlasov equation in EdS approximation (2.25) we find:

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}} \tag{4.3}$$

from which it follows:

$$F = 0 \implies \frac{df}{d\eta} = 0 \implies f(\eta, \mathbf{q}, \mathbf{p}) = f_{ini}(\eta_0, \mathbf{q}_0, \mathbf{p}_0) \tag{4.4}$$

4.0.1 Free Streaming

In free streaming approximation we ignore gravitational interactions, such that we set $\tilde{\Phi} = 0$ and thus $\nabla_{\mathbf{q}}\tilde{\Phi} = 0$:

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2}; \quad \frac{df}{ds} = 0} \tag{4.5}$$

$\implies s = \eta$ if $\eta(0) = 0$.

$$\mathbf{p}(\eta) = \mathbf{p}(\eta_0)\, e^{-\eta/2} = \mathbf{p}_0\, e^{-\eta/2} \tag{4.6}$$

$$\mathbf{q}(\eta) = \mathbf{q}_0 + \int_0^\eta \frac{\mathbf{p}(\eta')}{m}\, d\eta' = \mathbf{q}_0 + \frac{\mathbf{p}_0}{m}\int_0^\eta e^{-\eta'/2}\, d\eta' = \mathbf{q}_0 + 2\frac{\mathbf{p}_0}{m}\left[1 - e^{-\eta/2}\right] \tag{4.7}$$

In free streaming we read off the trajectories:

$$\left(\eta,\ \mathbf{q}_0 + 2\frac{\mathbf{p}_0}{m}\left[1 - e^{-\eta/2}\right],\ \mathbf{p}_0\, e^{-\eta/2}\right) \tag{4.8}$$

along which $f(\eta, \mathbf{q}(\eta), \mathbf{p}(\eta)) = f(0, \mathbf{q}_0, \mathbf{p}_0)$.

Other than defining a surface on which partial differential equations become ordinary differential equations, characteristics also hold physical meaning in our case: the characteristics define our particle trajectories. Additionally the method of characteristics will allow us to describe how the perturbations of f behave.

We start by introducing the comoving split similar to how we did in Chapter 3: $f = f_0 + \delta f$ Defining a background term as we did is also similar to Chapter 3, since we can interpret $\nabla\tilde{\Phi}$ as the gravitational operator. We also need to keep in mind the symmetries imposed on the comoving background:

- homogeneous in space ($\nabla_q f_0 = 0$)
- isotropic in momentum
- solves Vlasov eq. without gravity

It follows that

$$\frac{df_0}{d\eta} = 0 \implies \frac{\partial f_0}{\partial \eta} + \underbrace{\frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{q}} f_0}_{=0} - \frac{\mathbf{p}}{2}\cdot\nabla_{\mathbf{p}} f_0 = 0 \tag{4.9}$$

where we set the following as the specific solution for the free streaming regime

$$\frac{\partial f_0}{\partial \eta} - \frac{\mathbf{p}}{2}\cdot\nabla_{\mathbf{p}} f_0 = 0 \tag{4.10}$$

4.0.2 Zel'dovich "Free Streaming"

In this approach we look at the case where we say that the first order gravitational contribution is included into the background, so once again we have to come up with some notion of a background.
We first look at the general "free-streaming" case and see whether we can derive some notion of a background from it.
Vlasov:

$$\frac{\partial f}{\partial \eta} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}\right]\nabla_{\mathbf{p}} f = 0 \tag{4.11}$$

As I see it there is only one approach to this: we have to look at our analysis in Fourier space and just take our approach from there, where we stated:

$$f_Z = f_0 + \delta f^{(1)}; \quad \delta f_Z = \delta f^{(2)} + \delta f^{(3)} + \cdots \tag{4.12}$$

and f_Z is the function which solves

$$\frac{\partial f_Z}{\partial \eta} + \frac{\mathbf{p}}{m}\nabla_{\mathbf{q}} f_Z - \left[\frac{\mathbf{p}}{2} + m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}\right]\nabla_{\mathbf{p}} f_Z = 0 \tag{4.13}$$

from which we can extract the characteristics:

$$\boxed{\frac{d\eta}{ds} = 1; \quad \frac{d\mathbf{q}}{ds} = \frac{\mathbf{p}}{m}; \quad \frac{d\mathbf{p}}{ds} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)} \quad \text{and} \quad \frac{df}{d\eta} = 0} \tag{4.14}$$

In particular, with this choice of characteristics, we notice that the Zel'dovich background solution remains unchanged along the characteristics. This is a monumental result which will be the base of construction of f later on.

CHAPTER 5

Deviations from Zel'dovich Trajectories

Our next step is to get an expression for how the distribution function changes along the characteristics if we perturb the particle trajectories (characteristics).
Our final expression will look like

$$\begin{aligned} f(\mathbf{q},\mathbf{p},t) &= f\Big(\mathbf{q}\big(\mathbf{q}^{(i)},\mathbf{p}^{(i)},t\big),\ \mathbf{p}\big(\mathbf{q}^{(i)},\mathbf{p}^{(i)},t\big)\Big) \\ &= f\Big(\mathbf{q}_Z+\delta\mathbf{q},\ \mathbf{p}^{(i)}+\delta\mathbf{p}\Big) \\ &\approx f_{ini}\Big(\mathbf{q}_Z,\mathbf{p}^{(i)}\Big)+\partial_{\mathbf{q}}f_{ini}\cdot\delta\mathbf{q}+\partial_{\mathbf{p}}f_{ini}\cdot\delta\mathbf{p} \end{aligned} \tag{5.1}$$

where

$$\mathbf{q}=\mathbf{q}^{(i)}+\mathbf{D}+\mathbf{p}^{(i)}+\delta\mathbf{q}=\mathbf{q}_Z+\delta\mathbf{q} \tag{5.2}$$

$$\mathbf{p}=\mathbf{p}^{(i)}+\delta\mathbf{p} \qquad\qquad =\mathbf{p}^{(i)}+\delta\mathbf{p} \tag{5.3}$$

So for our purposes we need to find expressions for $\delta\mathbf{q}$ and $\delta\mathbf{p}$, since we already know f_0.
We will be using the EdS approximation of the Vlasov equation under new time $\eta=\log\Big(\frac{D(\tau)}{D^{(i)}}\Big)$, where τ is the conformal time.

$$\boxed{\frac{df}{d\eta}=\frac{\partial f}{\partial\eta}+\frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{q}}f-\left[\frac{\mathbf{p}}{2}+m\,\nabla_{\mathbf{q}}\tilde{\Phi}\right]\cdot\nabla_{\mathbf{p}}f=0} \tag{5.4}$$

5.1 Perturbed Paths

Goal: On the basis of the trajectories $\mathbf{q}_Z$ and $\mathbf{p}_Z$ we wish to evaluate the evolution of f along a path slightly perturbed from these.

We introduce: $\mathbf{q}(\eta) = \mathbf{q}_Z(\eta) + \delta\mathbf{q}(\eta)$; $\mathbf{p}(\eta) = \mathbf{p}_Z(\eta) + \delta\mathbf{p}(\eta)$.

$$\begin{aligned}
\implies \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}) &= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z + \delta\mathbf{q}) \\
&\approx \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \nabla_{\mathbf{q}}\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q} \\
&= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \frac{\partial^2\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)}{\partial_i\partial_j}\,\delta\mathbf{q} \qquad (5.1) \\
&= \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) + \mathbf{H}[\tilde{\Phi}^{(1)}]\delta\mathbf{q} \qquad (5.2)
\end{aligned}$$

Where $H[\tilde{\Phi}^{(1)}]$ denotes the Hessian of $\tilde{\Phi}$. From now on we will be referring to this Hessian as $H^{(1)}$ which naturally retains its time and positional dependence.

$$\text{note:} \quad \frac{d\mathbf{q}}{d\eta} = \frac{\mathbf{p}}{m} \implies \frac{d}{d\eta}\delta\mathbf{q} = \frac{\delta\mathbf{p}}{m} \qquad (5.3)$$

and

$$\frac{d\mathbf{p}}{d\eta} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}) \qquad (5.4)$$

$$\begin{aligned}
\implies \frac{d}{d\eta}\mathbf{p}_Z + \frac{d}{d\eta}\delta\mathbf{p} &= -\frac{\mathbf{p}_Z}{m} - \frac{\delta\mathbf{p}}{m} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z) - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q} \\
&\approx -\left[\frac{\mathbf{p}_Z}{m} + \nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}(\eta, \mathbf{q}_Z)\cdot m\right] - \left[\frac{\delta\mathbf{p}}{m} + m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}\right] \qquad (5.5)
\end{aligned}$$

we find the differential equation for the perturbed momentum:

$$\boxed{\implies \frac{\partial}{\partial\eta}\delta\mathbf{p} \approx -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}\cdot m} \qquad (5.6)$$

However, we notice the dependence on the perturbed comoving position. We can express this in terms of the perturbed momentum by solving the second equation and inserting.
We have: $\frac{d}{d\eta}\delta\mathbf{q} = \frac{\delta\mathbf{p}}{m} \implies \delta\mathbf{q} = \frac{\delta\mathbf{p}}{m}\eta + \delta\mathbf{q}_0$
which yields the closed differential equation for the momentum:

$$\begin{aligned}
\implies \frac{d}{d\eta}\delta\mathbf{p} &\approx -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\cdot m\left[\frac{\delta\mathbf{p}}{m}\eta + \delta\mathbf{q}_0\right] \\
&= -\frac{\delta\mathbf{p}}{m} - \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{p}\,\eta - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0 \\
&= -\delta\mathbf{p}\left[\frac{1}{m}\mathbf{I} + \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\eta\right] - m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0 \qquad (5.7)
\end{aligned}$$

5.1.1 Variation of Constants

We can finally solve these equations by making use of the variation of constants:

$$\frac{d}{d\eta}\delta\mathbf{p} = -\delta\mathbf{p}\underbrace{\left[\frac{1}{m}\mathbf{I} + \mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\eta\right]}_{=-a(\eta)} - \underbrace{m\,\mathbf{H}^{(1)}(\eta, \mathbf{q}_Z)\,\delta\mathbf{q}_0}_{=b(\eta)} \qquad (5.8)$$

$$\implies \delta\mathbf{p}(\eta) = \delta\mathbf{p}_H(\eta)\Big(\delta\mathbf{p}_0 + \int_0^\eta -m\mathbf{H}^{(1)}(\eta',\mathbf{q}_Z)\,\delta\mathbf{p}_H^{-1}\,\delta\mathbf{q}_0(\eta')\,d\eta'\Big) \tag{5.9}$$

$$\delta\mathbf{p}_H(\eta) = \exp\Big(\int_0^\eta -\Big[\frac{1}{m}\mathbf{I} + \mathbf{H}^{(1)}(\eta',\mathbf{q}_Z)\,\eta'\Big]\,d\eta'\Big) \tag{5.10}$$

Where $\delta\mathbf{p}_H$ denotes the homogeneous solution.

Our next hurdle lies in finding a computable expression for the Hessian $\mathbf{H}^{(1)}$. Fortunately the EdS approximation and hydrodynamical approaches let us solve for $\mathbf{H}^{(1)}$ rather elegantly:

$$\eta = \log\Big(\frac{D(\tau)}{D^{(i)}}\Big) \qquad \implies \qquad e^\eta = \frac{D(\tau)}{D^{(i)}} \tag{5.11}$$

$$\implies D(\tau) = e^\eta\,D^{(i)} \equiv D(\eta) \tag{5.12}$$

It should be noted that τ is the conformal time in this case.
From the linear growth of cosmological structure formation [1] we know:

$$\boxed{\delta = \delta_0\,D_+(a) = \delta_0\,D^{(i)}\,e^\eta = \delta_{\text{ini}}\,e^\eta} \tag{5.13}$$

In this ansatz we can replace using $\delta f^{(1)}$ and instead make use of $\delta^{(1)}$.
Next step is expressing $\tilde{\Phi}^{(1)}$ in terms of $\delta^{(1)} = \delta^{(1)}_{\text{ini}}\,e^\eta$:

$$\Delta\tilde{\Phi}^{(1)} = \frac{4\pi G\,\bar{\rho}\,a^2}{(Hf_+)^2}\,\delta^{(1)} \qquad G(\mathbf{q},\mathbf{q}') = -\frac{1}{4\pi|\mathbf{q}-\mathbf{q}'|} \tag{5.14}$$

$$\begin{aligned}\implies \tilde{\Phi}^{(1)} &= \frac{4\pi G\,\bar{\rho}\,a^2}{(Hf_+)^2}\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}(\mathbf{q}',\eta)\,d^3q' \\ &\equiv c\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}(\mathbf{q}',\eta)\,d^3q' \\ &= c\int G(\mathbf{q},\mathbf{q}')\,e^\eta\,\delta^{(1)}_{\text{ini}}(\mathbf{q}',\eta_0)\,d^3q' \\ &= e^\eta\,c\int G(\mathbf{q},\mathbf{q}')\,\delta^{(1)}_{\text{ini}}(\mathbf{q}',\eta_0)\,d^3q'\end{aligned} \tag{5.15}$$

$$\implies H^{(1)}_{ij}(\mathbf{q},\eta) = e^\eta\,c\int\frac{\partial^2 G(\mathbf{q},\mathbf{q}')}{\partial q_i\,\partial q_j}\,\delta^{(1)}_{\text{ini}}(\mathbf{q}',\eta_0)\,d^3q' \tag{5.16}$$

$$= e^\eta\,H^{(1)}_{\text{ini},\,ij}(\mathbf{q},\eta_0) \tag{5.17}$$

$$\boxed{H^{(1)}(\mathbf{q},\eta) = e^\eta\,H^{(1)}_{\text{ini}}(\mathbf{q},\eta_0)} \tag{5.18}$$

Inserting this result back into (5.10) we find

$$\delta\mathbf{p}_H(\eta) = \exp\Big(-\frac{\eta}{m}\mathbf{I}\Big)\exp\Big(-\int_0^\eta e^{\eta'}\,H^{(1)}_{\text{ini}}(\mathbf{q},\eta_0)\,\eta'\,d\eta'\Big) \tag{5.19}$$

$$= \exp\Big(-\frac{\eta}{m}\mathbf{I}\Big)\exp\Big(-H^{(1)}_{ini}\int_0^\eta e^{\eta'}\eta'\,d\eta'\Big) \tag{5.20}$$

$$= \exp\Big(-\frac{\eta}{m}\mathbf{I}\Big)\exp\Big(-H^{(1)}_{ini}\,[\mathrm{e}^\eta(\eta-1)+1]\Big) \tag{5.21}$$

$$\boxed{\delta\mathbf{p}(\eta) = \delta\mathbf{p}_H(\eta)\Big(\delta\mathbf{p}_0 + -m\, H^{(1)}_{\text{ini}}(\mathbf{q},\eta_0)\int_0^\eta e^{\eta'}\ \delta\mathbf{p}_H^{-1}\delta\mathbf{q}_0\, d\eta'\Big)} \tag{5.22}$$

$$\boxed{\delta\mathbf{q} = \frac{\delta\mathbf{p}(\eta)}{m}\,\eta + \delta\mathbf{q}_0} \tag{5.23}$$

These are the last step in determining the evolution of f along the perturbed characteristics. We remind:

$$f(\mathbf{q},\mathbf{p},\eta) \approx f_{ini}(\mathbf{q}_Z,\mathbf{p}^{(i)}) + \partial_{\mathbf{q}} f_{ini}\cdot\delta\mathbf{q} + \partial_{\mathbf{p}} f_{ini}\cdot\delta\mathbf{p} \tag{5.24}$$

5.1.2 Some remarks on this approach

In spite of the correct formulation with respect to the Zel'dovich frames one might notice that we have restricted ourselves to the first order gravitational contributions, such that even if we are in fact analysing paths deviating from the Zel'dovich trajectories we can only ever hope to analyse dynamics which arise from the first order Hessian $\mathbf{H^{(1)}}$ contributions.
If we want to take a look at what dynamics might be produced from higher order gravitational contributions, we must proceed with a Born approximation.

5.2 Born Approximation for Perturbed Trajectories

In this approximation we will attempt to get to a tractable expression for the perturbed trajectories. We will compute the effect of a perturbation by evaluating it along the unperturbed solution (Zel'dovich trajectories).
We start off assuming that the change in position is given by a velocity field and that the change in momentum is given by a force field:

$$\dot{\mathbf{q}} = \mathbf{v}; \quad \dot{\mathbf{p}} = \mathbf{F}(\mathbf{q}, \mathbf{p}, \eta) \quad \Longrightarrow \quad \dot{\mathbf{q}} = \frac{\mathbf{p}}{m}; \quad \dot{\mathbf{p}} = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi} \tag{5.1}$$

We get the right hand side from the characteristics of the Einstein–de Sitter Universe, where time is $\eta = \log\left(\frac{D(\tau)}{D^{(i)}}\right)$.

$$\tilde{\Phi} = \tilde{\Phi}^{(1)} + \varepsilon\,\delta\tilde{\Phi} \quad \Longrightarrow \quad \mathbf{F}_0 = -\frac{\mathbf{p}}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)}, \quad \mathbf{F}_1(\mathbf{q}, \eta) = -m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi} \tag{5.2}$$

$$\Longrightarrow \mathbf{F} = \mathbf{F}_0 + \varepsilon\,\mathbf{F}_1 \tag{5.3}$$

$$\Longrightarrow \dot{\mathbf{q}}_Z = \frac{\mathbf{p}_Z}{m}; \quad \dot{\mathbf{p}}_Z = \mathbf{F}_0(\mathbf{q}_Z, \mathbf{p}_Z, \eta) \tag{5.4}$$

$$\mathbf{q} = \mathbf{q}_Z + \varepsilon\,\delta\mathbf{q} \qquad \mathbf{p} = \mathbf{p}_Z + \varepsilon\,\delta\mathbf{p} \tag{5.5}$$

$$\begin{aligned}
\dot{\mathbf{p}} = \dot{\mathbf{p}}_Z + \varepsilon\,\delta\dot{\mathbf{p}} &= \mathbf{F}_0(\mathbf{q}_Z + \varepsilon\,\delta\mathbf{q},\ \mathbf{p}_Z + \varepsilon\,\delta\mathbf{p},\ \eta) + \varepsilon\,\mathbf{F}_1(\mathbf{q}_Z + \varepsilon\,\delta\mathbf{q},\ \eta) \\
&= \mathbf{F}_0(\mathbf{q}_Z, \mathbf{p}_Z) + \varepsilon\left[\left.\frac{\partial\mathbf{F}_0}{\partial\mathbf{q}}\right|_Z \delta\mathbf{q} + \left.\frac{\partial\mathbf{F}_0}{\partial\mathbf{p}}\right|_Z \delta\mathbf{p}\right] + \varepsilon\,\mathbf{F}_1(\mathbf{q}_Z, \eta) + \mathcal{O}(\varepsilon^2) \\
&= -\frac{\mathbf{p}_Z}{2} - m\,\nabla_{\mathbf{q}}\tilde{\Phi}^{(1)} + \varepsilon\left[-m \underbrace{\frac{\partial^2\tilde{\Phi}^{(1)}}{\partial q_i\,\partial q_j}}_{=H[\tilde{\Phi}^{(1)}]=H^{(1)}} \delta\mathbf{q} - \frac{1}{2}\delta\mathbf{p} + \mathbf{F}_1(\mathbf{q}_Z, \eta)\right] + \mathcal{O}(\varepsilon^2)
\end{aligned} \tag{5.6}$$

$$\Longrightarrow \delta\dot{\mathbf{p}} = -m\,H^{(1)}\delta\mathbf{q} - \frac{1}{2}\delta\mathbf{p} + \mathbf{F}_1(\mathbf{q}_Z, \eta); \qquad \delta\dot{\mathbf{q}} = \frac{\delta\mathbf{p}}{m} \tag{5.7}$$

$$\boxed{\frac{d}{d\eta}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = \underbrace{\begin{pmatrix}0 & \frac{1}{m}\mathbf{I}\\ -m\,H^{(1)} & -\frac{1}{2}\mathbf{I}\end{pmatrix}}_{\equiv M(\eta)}\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} + \begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z, \eta)\end{pmatrix}} \tag{5.8}$$

This differential equation gets solved by a Volterra type integral:

$$\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix} = \Psi(\eta, \eta_0)\begin{pmatrix}\delta\mathbf{q}_0\\ \delta\mathbf{p}_0\end{pmatrix} + \int_{\eta_0}^{\eta}\Psi(\eta, s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s), s)\end{pmatrix} ds \tag{5.9}$$

$$\boxed{\approx \int_{\eta_0}^{\eta}\Psi(\eta, s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s), s)\end{pmatrix} ds} \tag{5.10}$$

where we assume vanishing deviations from the Zel'dovich trajectories at the initial time in the last step.
We define the propagator $\Psi(\eta, \eta_0)$ as:

$$\partial_\eta \, \Psi(\eta, \eta') = M(\eta') \, \Psi(\eta, \eta'); \qquad \Psi(\eta', \eta') = \mathbf{I} \tag{5.11}$$

$$\implies \Psi(\eta, \eta') = \exp\Big(\int_{\eta'}^{\eta} M(\eta^*) \, d\eta^* \Big) \quad \longleftarrow \quad \text{series expansion} \tag{5.12}$$

The next step is to find an appropriate truncation of this series expansion. We drop any contributions which are squared in the Hessian $H^{(1)}$ (or higher), since the Hessian depends component-wise on $\delta f^{(1)}$.

5.2.1 Decomposition of the Propagator

To proceed we must decompose $M(\eta)$:

$$M(\eta) = \begin{pmatrix} 0 & \frac{1}{m}\mathbf{I} \\ -m\, H^{(1)} & -\frac{1}{2}\mathbf{I} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & \frac{1}{m}\mathbf{I} \\ 0 & -\frac{1}{2}\mathbf{I} \end{pmatrix}}_{\equiv M_0} + \underbrace{\begin{pmatrix} 0 & 0 \\ -m\, H^{(1)} & 0 \end{pmatrix}}_{\equiv M_1(\eta)} \tag{5.13}$$

5.2.2 The Homogeneous Solution Ψ

$$\frac{d}{d\eta} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} = M(\eta) \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \quad \iff \quad \frac{d}{d\eta} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} = (M_0 + M_1) \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \tag{5.14}$$

By inspection we find the free evolution propagator:

$$\Psi_0 = \exp\Big(M_0(\eta - \eta_0) \Big) \tag{5.15}$$

What about the rest of the evolution? (i.e. non-free evo.)

$$\begin{pmatrix} \delta\tilde{\mathbf{q}} \\ \delta\tilde{\mathbf{p}} \end{pmatrix} = \Psi_0^{-1} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} = e^{-M_0(\eta - \eta_0)} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \tag{5.16}$$

$$\begin{aligned} \implies \frac{d}{d\eta} \begin{pmatrix} \delta\tilde{\mathbf{q}} \\ \delta\tilde{\mathbf{p}} \end{pmatrix} &= \frac{d}{d\eta} \left[\Psi_0^{-1} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \right] = \left[\frac{d}{d\eta} \Psi_0^{-1} \right] \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} + \Psi_0^{-1} \frac{d}{d\eta} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \\ &= -M_0 \, \Psi_0^{-1} \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} + \Psi_0^{-1} (M_0 + M_1) \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \\ &= \Psi_0^{-1} \, M_1 \begin{pmatrix} \delta\mathbf{q} \\ \delta\mathbf{p} \end{pmatrix} \\ &= \Psi_0^{-1} \, M_1 \, \Psi_0 \begin{pmatrix} \delta\tilde{\mathbf{q}} \\ \delta\tilde{\mathbf{p}} \end{pmatrix} \end{aligned} \tag{5.17}$$

$$\implies \frac{d}{d\eta} \begin{pmatrix} \delta\tilde{\mathbf{q}} \\ \delta\tilde{\mathbf{p}} \end{pmatrix} = \Psi_0^{-1} \, M_1(\eta) \, \Psi_0 \begin{pmatrix} \delta\tilde{\mathbf{q}} \\ \delta\tilde{\mathbf{p}} \end{pmatrix} \tag{5.18}$$

5.2.3 The Non-Free Propagator

The "non-free" propagator follows as:

$$\boxed{\Psi_1 = \exp\left(\int_{\eta_0}^{\eta} \Psi_0^{-1}\, M_1(\eta')\, \Psi_0\, d\eta'\right)} \tag{5.19}$$

$$\implies \Psi(\eta,\eta_0) = \Psi_0(\eta,\eta_0)\, \Psi_1(\eta,\eta_0) \qquad \text{let } \tau = \eta - \eta_0 \tag{5.20}$$

$$= e^{M_0\tau} \exp\left(\int_{\eta_0}^{\eta} \underbrace{e^{-M_0\tau(\eta')}\, M_1(\eta')\, e^{M_0\tau(\eta')}}_{\Psi_0^{-1}\, M_1(\eta')\, \Psi_0}\, d\eta'\right) \tag{5.21}$$

We continue by searching for a truncation of this operator, as the argument of the second exponent depends on the Hessian.
We first take a look at $e^{M_0\tau}$ and $e^{-M_0\tau}$:

$$e^{M_0\tau} = \sum_{n=0}^{\infty} \frac{(M_0\tau)^n}{n!} = \mathbf{I} + \sum_{n=1}^{\infty} \frac{M_0^n\, \tau^n}{n!} \tag{5.22}$$

where it can be shown that:

$$M_0^n = \begin{pmatrix} 0 & \frac{1}{(-2)^{n-1}} \frac{1}{m} \mathbf{I} \\ 0 & \frac{1}{(-2)^n} \mathbf{I} \end{pmatrix} \tag{5.23}$$

Top right component:

$$\begin{aligned} \frac{1}{m} \sum_{n=1}^{\infty} \frac{1}{(-2)^{n-1}} \cdot \frac{1}{n!}\, \tau^n &= \frac{1}{m} \sum_{n=1}^{\infty} \frac{-2}{(-2)^n} \cdot \frac{1}{n!}\, \tau^n \\ &= -\frac{2}{m} \sum_{n=1}^{\infty} \left(-\frac{\tau}{2}\right)^n \cdot \frac{1}{n!} \\ &= -\frac{2}{m} \left(e^{-\tau/2} - 1\right) \end{aligned} \tag{5.24}$$

Bottom right component:

$$\sum_{n=1}^{\infty} \frac{\tau^n}{(-2)^n} \cdot \frac{1}{n!} = e^{-\tau/2} - 1 \tag{5.25}$$

$$\implies \Psi_0 = e^{M_0\tau} = \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1 - e^{-\tau/2})\, \mathbf{I} \\ 0 & e^{-\tau/2}\, \mathbf{I} \end{pmatrix} \tag{5.26}$$

$$\implies \Psi_0^{-1} = e^{-M_0\tau} = \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1 - e^{\tau/2})\, \mathbf{I} \\ 0 & e^{\tau/2}\, \mathbf{I} \end{pmatrix} \tag{5.27}$$

$$\begin{aligned} \implies \tilde{M}_1(\eta) &= \Psi_0^{-1}\, M_1\, \Psi_0 \\ &= \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1 - e^{\tau/2})\, \mathbf{I} \\ 0 & e^{\tau/2}\, \mathbf{I} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -m\, H^{(1)} & 0 \end{pmatrix} \begin{pmatrix} \mathbf{I} & \frac{2}{m}(1 - e^{-\tau/2})\, \mathbf{I} \\ 0 & e^{-\tau/2}\, \mathbf{I} \end{pmatrix} \\ &= \begin{pmatrix} 2(e^{\tau/2} - 1)\, H^{(1)} & -\frac{4}{m}(1 - e^{\tau/2})(1 - e^{-\tau/2})\, H^{(1)} \\ m\, e^{\tau/2}\, H^{(1)} & 2\, e^{\tau/2}(1 - e^{-\tau/2})\, H^{(1)} \end{pmatrix} \end{aligned} \tag{5.28}$$

We notice that every component in the block form depends on the Hessian and thus on $\delta f^{(1)}$. We can now finally truncate at second order.

$$\begin{aligned}
\implies \Psi(\eta,\eta_0) &= \Psi_0(\eta,\eta_0)\,\Psi_1(\eta,\eta_0) \\
&= e^{M_0\tau}\exp\Big(\int_{\eta_0}^{\eta}\Psi_0^{-1}\,M_1\,\Psi_0\,d\eta'\Big) \\
&= e^{M_0\tau}\exp\Big(\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\Big) \\
&\approx e^{M_0\tau}\Big\{\mathbf{I}+\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\Big\}+\mathcal{O}\Big((H^{(1)})^2\Big)
\end{aligned} \tag{5.29}$$

$$\boxed{\implies \Psi(\eta,\eta_0)\approx e^{M_0\tau}\Big\{\mathbf{I}+\int_{\eta_0}^{\eta}\tilde{M}_1(\eta')\,d\eta'\Big\}} \tag{5.30}$$

Taking a look at (5.28) we notice that each component depends on $H^{(1)}$ and according to (5.18) this results in $\tilde{M}_1$ being linear in e^{η}, making (5.30) tractable.

5.2.4 Resolving the Nested Integral

We can now return to our suggested solution (5.10):

$$\begin{pmatrix}\delta\mathbf{q}\\ \delta\mathbf{p}\end{pmatrix}\approx\int_{\eta_0}^{\eta}\Psi(\eta,s)\begin{pmatrix}0\\ \mathbf{F}_1(\mathbf{q}_Z(s),s)\end{pmatrix}ds \tag{5.31}$$

While $\Psi(\eta,s)$ is still an integral, we know it to be tractable, since it depends on known quantities. The only issue left to deal with is $\mathbf{F}_1$, as it depends on $\delta\tilde{\Phi}$, which we can treat analogously to the Hessian as done in (5.18).

$$\mathbf{F}_1(\mathbf{q}_Z(s),s)=-m\,\nabla_{\mathbf{q}}\delta\tilde{\Phi}(\mathbf{q},s)\Big|_{\mathbf{q}=\mathbf{q}_Z(s)} \tag{5.32}$$

$$=-m\,e^{s}\int\nabla_{\mathbf{q}}G(\mathbf{q},\mathbf{q}')\,\delta_{\mathrm{ini}}^{(n\geq 2)}(\mathbf{q}',\eta_0)\,d^3q'\Big|_{\mathbf{q}=\mathbf{q}_Z(s)} \tag{5.33}$$

which is also tractable.
It should be noted that for the derivation of Ψ we truncate the expression at the second order to retain the Zel'dovich nature of the propagator; yet our final expressions for the perturbed position and momentum depend on F_1 which notably depends on perturbations of order 2 and higher. One might therefore question the validity of the prior truncation; however this procedure retains its validity due to the nature of Born approximations, since we evaluate the effects of perturbations along unperturbed paths.

CHAPTER 6

Initial Conditions and the Gaussian Closure

The final step in finding the approximate solution for f lies in finding an initial condition $f_Z = f_{ini}$, which notably remains unchanged along characteristics according to previous calculations. Since our approximate solution f not only depends on f_0 but also on $\partial_q f_0$ as well as $\partial_p f_0$ we must require it to be differentiable in both position and momentum. The SSA solution proves to be unusable due to the non-differentiable nature of the Dirac Delta function which appears; however we can make use of the expression that includes cumulants up to the second order (cf. Eq. (2.20) in Sec. 2.4). Starting from the general form above and imposing *isotropy*,

$$\sigma_{ij} = \bar{\sigma}^2\,\delta_{ij} \quad \Rightarrow \quad (\sigma^{-1})_{ij} = \frac{1}{\bar{\sigma}^2}\,\delta_{ij}, \tag{6.1}$$

so the distribution simplifies to

$$f_2(\mathbf{q},\mathbf{p},\tau) = \frac{n(\mathbf{q},\tau)}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{(\mathbf{p}-am\,\mathbf{u})^2}{2\bar{\sigma}^2}\right). \tag{6.2}$$

At the initial time $\eta = 0$ we set

$$n(\mathbf{q}^{(i)}) = \frac{\bar{\rho}}{m}\Big(1+\delta_{ini}(\mathbf{q}^{(i)})\Big), \qquad am\,\mathbf{u}_{ini} = \nabla_q\psi_{ini} \tag{6.3}$$

where ψ_{ini} is the initial velocity potential. This gives

$$f_s(\mathbf{q},\mathbf{p},\tau) = \frac{\bar{\rho}\Big(1+\delta_{ini}(\mathbf{q}^{(i)})\Big)}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{\Big(\mathbf{p}-\nabla_q\psi_{ini}\Big)^2}{2\bar{\sigma}^2}\right). \tag{6.4}$$

The density perturbation and the velocity potential are related by [1]

$$\rho_{ini} = \nabla_q\psi_{ini}(\mathbf{q}^{(i)}) \;\Rightarrow\; \delta_{ini} = -\nabla_q^2\psi_{ini}, \qquad \Rightarrow\; \psi_{ini} = -\frac{\phi_{ini}}{A_\phi}, \quad \text{where } \nabla^2\phi_{ini} = A_\phi\,\delta_{ini}. \tag{6.5}$$

Since $\psi^{(i)}$ is proportional to $\phi^{(i)}$, to maintain first order consistency we need to truncate f_0 at first order in gravitational perturbations. Expanding the Gaussian:

$$\begin{aligned}
\exp\left(-\frac{(\mathbf{p}-\nabla_q\psi_{ini})^2}{2\bar{\sigma}^2}\right) &= \exp\left(-\frac{p^2 - 2\,\mathbf{p}\cdot\nabla_q\psi_{ini} + |\nabla_q\psi_{ini}|^2}{2\bar{\sigma}^2}\right) \\
&\approx \exp\left(-\frac{p^2}{2\bar{\sigma}^2} + \frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right) \qquad \left(\text{drop } |\nabla_q\psi_{ini}|^2 \sim \mathcal{O}(\phi_{ini}^2)\right) \\
&= \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\exp\left(\frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right) \\
&\approx \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right), \qquad (\text{exp series truncated at } \mathcal{O}(\phi_{ini}^2))
\end{aligned} \tag{6.6}$$

yielding the initial distribution linearised in gravitational perturbations:

$$f_0(\mathbf{q},\mathbf{p},\tau) = \frac{\bar{\rho}\left(1+\delta_{ini}(\mathbf{q}^{(i)})\right)}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right) \tag{6.7}$$

To first order this is

$$\boxed{f_0(\mathbf{q},\mathbf{p},\tau) \approx \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1 + \delta_{ini}(\mathbf{q}^{(i)}) + \frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right)} \tag{6.8}$$

6.1 Derivatives of f_0

The spatial and momentum derivatives needed for the Vlasov equation are:

$$\partial_{q_i} f_0 = \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(\partial_{q_i}\delta^{(i)}(\mathbf{q}^{(i)}) + \frac{p_j\,\partial_{q_i}\partial_{q_j}\psi_{ini}}{\bar{\sigma}^2}\right), \tag{6.1}$$

and, using the product rule on $f_0 = \frac{\bar{\rho}(1+\delta_{ini})}{(2\pi\bar{\sigma}^2)^{3/2}}e^{-p^2/2\bar{\sigma}^2}(1+\mathbf{p}\cdot\nabla_q\psi^{(i)}/\bar{\sigma}^2)$:

$$\partial_{p_i} f_0 = \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}}\exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left[\frac{\partial_{q_i}\psi_{ini}}{\bar{\sigma}^2} - \frac{p_i}{\bar{\sigma}^2}\left(1 + \delta_{ini}(\mathbf{q}^{(i)}) + \frac{\mathbf{p}\cdot\nabla_q\psi_{ini}}{\bar{\sigma}^2}\right)\right]. \tag{6.2}$$

The only thing left to do is to actually evaluate δq and δp; for this we need to evaluate the Zel'dovich propagator Ψ as well as the integral in Eq. (5.10). If everything is done correctly, we

are left with an expression for f:

$$
\begin{aligned}
f \approx\ & \frac{\bar{\rho}(1+\delta_{ini}^{(1)}(q^{(i)}))}{(2\pi\bar{\sigma})^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(1+\frac{p\cdot\nabla_q\psi_{ini}^{(1)}}{\bar{\sigma}^2}\right) \\
&+ \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(\nabla_q\delta_{ini}^{(1)}(q^{(i)})+\frac{p\,\mathbf{H}[\psi_{ini}^{(1)}]}{\bar{\sigma}^2}\right) \\
&\quad\cdot\left\{\frac{4}{m}\left(\frac{1}{9}e^{3\eta}-\frac{16}{9}e^{3\eta/2}-\frac{9}{30}e^{2\eta}+\frac{1}{6}+\frac{2}{15}e^{-\eta/2}+\frac{11}{3}e^{\eta}-2e^{\eta/2}\right)H_{\text{ini}}^{(1)}+\frac{4}{m}\left(\frac{2}{3}e^{\eta}-2+\frac{4}{3}e^{-\eta/2}\right)\mathbb{I}\right\}F_{\text{ini}} \\
&+ \frac{\bar{\rho}}{(2\pi\bar{\sigma}^2)^{3/2}} \exp\left(-\frac{p^2}{2\bar{\sigma}^2}\right)\left(\frac{\nabla_q\psi_{ini}^{(1)}}{\bar{\sigma}^2}-\frac{p}{\bar{\sigma}^2}\left(1+\delta_{ini}^{(1)}(q^{(i)})\right)+\frac{p\cdot\nabla_q\psi_{ini}^{(1)}}{\bar{\sigma}^2}\right) \\
&\quad\cdot\left\{2\left[\frac{12}{15}e^{2\eta}-\frac{4}{3}e^{\eta}-\frac{2}{15}e^{-\eta/2}+\frac{2}{3}e^{\eta/2}\right]H_{\text{ini}}^{(1)}+\left(\frac{2}{3}e^{\eta}-\frac{2}{3}e^{-\eta/2}\right)\mathbb{I}\right\}F_{\text{ini}}
\end{aligned}
\tag{6.3}
$$

We can go on to recover results for perfectly cold initial conditions by taking the limit $\bar{\sigma}\to 0$, which results in:

$$
\begin{aligned}
f \approx\ & \bar{\rho}\delta^{(3)}(p)+\bar{\rho}\delta^{(3)}(p)\nabla_q\delta_{ini}(q^{(i)}) \\
&\cdot\left\{\frac{4}{m}\left(\frac{1}{9}e^{3\eta}-\frac{16}{9}e^{3\eta/2}-\frac{9}{30}e^{2\eta}+\frac{1}{6}+\frac{2}{15}e^{-\eta/2}+\frac{11}{3}e^{\eta}-2e^{\eta/2}\right)H_{\text{ini}}^{(1)}+\frac{4}{m}\left(\frac{2}{3}e^{\eta}-2+\frac{4}{3}e^{-\eta/2}\right)\mathbb{I}\right\}F_{\text{ini}}
\end{aligned}
\tag{6.4}
$$

we notice in particular that the entire third term regarding the change in momentum drops out in this limit. This makes sense physically, since in this limit the momentum distribution approaches a delta distribution around the origin.

CHAPTER 7

Conclusion

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